

Learning Predator–Prey Interactions from Real-World Animal Behavior via Adversarial Imitation Learning

Abstract—Understanding how local interactions give rise to coordinated collective behavior is important for both biological research and decentralized robotic systems. Traditional models rely largely on hand-crafted interaction rules, while data-driven approaches have mostly focused on homogeneous groups or synthetic demonstrations. Here, we introduce a co-evolutionary generative adversarial imitation learning framework for heterogeneous predator–prey behavior. Separate predator and prey policies combine pairwise interaction and attention networks with role-specific discriminators, allowing both roles to learn distinct strategies through reciprocal interactions. We first validate the framework on controlled predator–prey trajectories generated by an extended Couzin model and then apply it to trajectories reconstructed from biological recordings. In the synthetic setting, the learned policies closely reproduce predator proximity and preserve the direction of several behavioral responses, although pursuit and escape alignment remain weaker than in the expert. On biological data, overall collective motion is captured more closely than directional pursuit and escape responses. These results demonstrate the feasibility of learning role-specific heterogeneous collective behavior from biological observations while highlighting tightly coupled inter-species interactions as a key challenge for imitation learning. Such policies could ultimately support heterogeneous multi-robot systems with asymmetric roles without relying on hand-crafted interaction rules.

I. INTRODUCTION

Collective behavior is a striking phenomenon in nature, observed in bird flocks, fish schools, and the foraging trails of ants [1], [2], [3]. Interactions among individuals can give rise to self-organized collective order, allowing groups to move as coherent units and change direction and shape without centralized or external coordination [1], [2], [4]. Understanding how such behavior emerges is important not only for biology but also for engineering, as the principles underlying natural collective motion have inspired decentralized control strategies for robotic swarms [5], [6].

Early computational models showed that coordinated collective motion can emerge from simple local interaction rules. Reynolds’ Boids model [2] reproduced flocking through separation, alignment, and cohesion, while Couzin et al. [1] described fish schools using concentric zones of repulsion, orientation, and attraction. These agent-based models are simple, interpretable, and remarkably successful, but they rely on assumed interaction rules rather than measured ones. The form of the interactions is specified a priori, and model parameters are adjusted until the simulated groups reproduce macroscopic properties such as polarization or group shape. Such agreement, however, does not uniquely validate the underlying behavioral rules, because different interaction mechanisms can generate similar collective pat-

terns [7]. Experimental observations further show that real animals do not interact according to fixed spatial zones: the influence of neighbors depends on both spatial configuration and behavioral context [8]. Extending rule-based models to capture these effects typically requires introducing additional hand-crafted rules for each newly observed behavior, making the models increasingly difficult to generalize. Data-driven approaches take the opposite strategy. Rather than prescribing interaction rules, they infer individual decision-making directly from behavioral observations.

Imitation learning provides a natural framework for learning collective behavior from demonstrations and encompasses approaches such as behavioral cloning (BC), inverse reinforcement learning (IRL), and adversarial imitation learning (AIL) [9]. BC directly predicts actions from observations. For example, Heras et al. [8] use an attention-based interaction network to predict zebrafish turning decisions from a weighted aggregation of neighboring fish, achieving 85% accuracy. Their model further shows that a fish attends roughly 17 neighbors, although its behavior is largely determined by about 5, and that this influential subset changes on a sub-second timescale. However, because BC is trained only on states the demonstrator visits, prediction errors accumulate during autonomous rollouts [10]. IRL instead seeks to recover a reward function that explains the demonstrated behavior, from which an imitation policy can subsequently be learned [11]. Schafer et al. [12] apply Bayesian IRL to tracked fish trajectories, while Yu et al. [13] jointly learn a shared reward function and policy from Vicsek-model demonstrations. AIL avoids explicitly recovering a reward function and instead trains a policy adversarially against a discriminator that distinguishes generated behavior from expert demonstrations [14]. Building on this approach, Yapei Wu et al. [6] combine AIL with an attention-based interaction network to reproduce milling, schooling, and swarming from Couzin-model demonstrations, then transfer the learned policy to a swarm of 20 wheeled robots. Yifan Wu et al. [15] similarly adversarially train one policy for all fish to match video-derived latent motion encoded by a masked autoencoder.

These approaches, however, largely rely on a single shared policy across individuals, an assumption that is poorly suited to predator–prey systems in which the two roles pursue opposing objectives and exhibit asymmetric pursuit and evasion strategies [16]. Here, we learn separate predator and prey policies directly from biological demonstrations, enabling us to model the two roles separately and evaluate how well their coupled dynamics can be recovered. We pro-

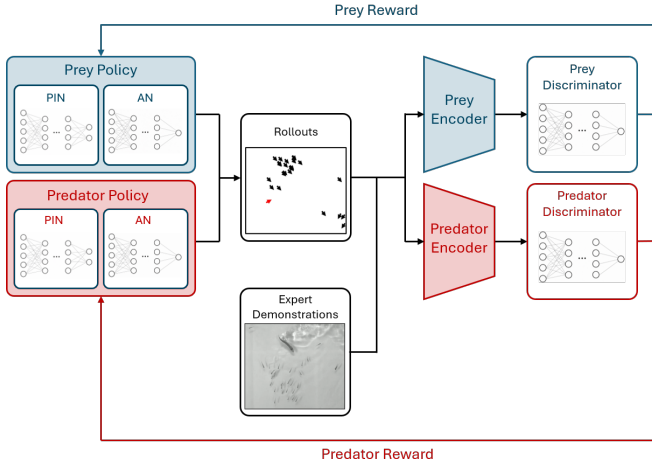


Fig. 1: Overview of the role-specific generative adversarial imitation learning (GAIL) framework. The predator policy and the policy shared by all prey combine a pairwise interaction network (PIN) with an attention network (AN). Joint policy execution produces predator–prey rollouts, which, together with role-matched expert demonstrations, are encoded into latent transitions and evaluated by separate predator and prey discriminators. Each discriminator provides a role-specific imitation reward.

pose two policies that are trained jointly through alternating optimization with an evolutionary strategy, allowing them to adapt to one another while remaining stable during learning. We first validate the framework on synthetic predator–prey demonstrations generated with the Couzin model and then apply it to real biological predator–prey recordings. We make three main contributions: (i) a role-asymmetric adversarial imitation learning framework with separate policies and discriminators for predators and prey; (ii) the application of alternating co-training to predator–prey learning with conflicting role-specific objectives; and (iii) validation on both synthetic and biological predator–prey datasets, demonstrating that the learned policies capture characteristic features of heterogeneous predator–prey behavior.

II. MULTI-SPECIES BEHAVIORAL MODELING FRAMEWORK

We propose a co-evolutionary generative adversarial imitation learning (GAIL) framework in which separate predator and prey policies interact in a shared environment (Fig. 1). One policy models predator behavior, while a parameter-shared policy controls all prey individuals. The two policies generate rollouts that are transformed into the same state representations as their corresponding expert demonstrations. Separate transition encoders and discriminators then evaluate whether predator and prey transitions match their respective expert behaviors. The resulting discriminator feedback is used to alternately optimize the two policies, allowing them to co-adapt through their interactions in the shared environment.

A. Role-specific Policy Architecture

Following [6], [8], we describe the predator and prey policies by $\pi_{P,\phi,\varphi}$ and $\pi_{Y,\phi,\varphi}$, respectively, where ϕ represents a pairwise interaction network (PIN) and φ an attention network (AN). This decomposition separates neighbor-specific responses from their aggregation, preserving interpretability while allowing the policy to scale with the number of neighbors. Both networks consist of three fully connected hidden layers with 128, 64, and 32 units and ReLU activations. For each focal–neighbor pair, the PIN processes the relative position and velocity features introduced in Section III-C. For the prey policy, the input additionally includes a binary predator flag, allowing the policy to distinguish the predator from conspecific neighbors. The PIN outputs mean μ_i and standard deviation σ_i of a Gaussian distribution $\mathcal{N}(\mu_i, \sigma_i)$ from which a neighbor-specific turning response is sampled:

$$a_i \sim \mathcal{N}(\mu_i, \sigma_i). \quad (1)$$

This stochastic formulation introduces behavioral variability instead of producing identical reactions under identical local conditions. The resulting sample diversity further helps mitigate the risk of mode collapse.

In parallel, the AN assigns an attention weight ω_i to each focal–neighbor interaction, representing the relative influence of neighbor i on the focal individual’s decision. The final policy output is obtained by aggregating the neighbor-specific responses through a normalized weighted sum:

$$a = \sum_{i \in \mathcal{V}} a_i \frac{\omega_i}{\sum_{j \in \mathcal{V}} \omega_j}, \quad (2)$$

where $\mathcal{V}$ denotes the set of valid neighbors and assuming that at least one valid neighbor exists.

The same aggregation principle is applied to both roles. For the prey policy, the predator is not handled through a separate aggregation rule. Instead, the predator flag allows the policy to learn role-specific responses directly from the demonstrations. The aggregated output defines the per-step heading change of the focal individual.

B. Latent Transition Encoding and Discrimination

Following Yifan Wu et al. [15], we use separate transition encoders, $E_{P,\psi}$ and $E_{Y,\psi}$, for predator and prey trajectories, respectively, to avoid discriminator inputs that scale with sequence length, agent number, neighborhood size, and feature dimension. For each agent and time step, neighbor-level information is first aggregated and mapped to a latent agent state z_t . The transition between two consecutive states is represented as:

$$e_t = [z_t, \Delta z_t], \quad \Delta z_t = z_{t+1} - z_t, \quad (3)$$

where e_t denotes the latent transition representation at time step t and Δz_t the change in latent state between consecutive time steps.

We pretrain encoders separately on the corresponding expert trajectories using Variance–Invariance–Covariance Regularization [17]. The self-supervised objective encourages

independently perturbed versions of the same trajectory sequence, generated through noise and masking, to obtain similar representations while preventing representational collapse. After pretraining, the projectors are discarded and the encoder parameters are frozen. Consequently, both expert and policy-generated trajectories are encoded within the same fixed latent space during adversarial training.

The resulting transition embeddings are evaluated by separate role-specific discriminators $D_{P,\beta}$ and $D_{Y,\beta}$. Each discriminator acts as a Wasserstein critic, assigning higher scores to transitions that resemble the corresponding expert behavior and lower scores to transitions that deviate from it [18]. A gradient penalty enforces the Lipschitz constraint and stabilizes discriminator training [19]. The resulting score difference between expert and policy-generated transitions provides the role-specific feedback used to optimize the predator and prey policies.

C. Co-Evolutionary Training

The predator and prey policies are initialized by BC [14], where the expected action is fitted to expert heading changes using a mean squared error objective. During each training generation, exponential moving average (EMA) copies [20] of both policies are jointly rolled out in the shared two-dimensional environment. Initial states are sampled from the expert demonstrations $\tau_{E,P}$ and $\tau_{E,Y}$. Each policy controls the per-step heading change of its respective agents, while movement speed is fixed at 10 pixels per step. This simplifying assumption of time-invariant speed reduces the action space and focuses learning on directional interactions, which is advantageous given the limited number of observed attack events. However, it prevents the policies from capturing role-specific speeds, acceleration, and burst responses (see future work in Sec. V). The resulting trajectories are transformed into the same state and transition representations as the expert demonstrations.

The role-specific discriminators are trained on encoded expert and generated transitions using Root Mean Square Propagation (RMSProp) [21], with multiple discriminator updates performed before each policy update. The predator and prey policies are then optimized using OpenAI Evolution Strategies (ES) [22]. Mirrored perturbations $\theta + \sigma\epsilon_i$ and $\theta - \sigma\epsilon_i$ are evaluated from identical initial states, and their discriminator-based returns determine the parameter update. Mirrored sampling and rank normalization are applied as by Yapei Wu et al. [6].

The predator and prey policies are optimized alternately (Algorithm 1), so that each updated policy changes the interaction environment encountered by the other in subsequent rollouts. Training is further stabilized using ES update clipping, early discriminator input noise, and gradual decay γ of the learning rates λ_P, λ_Y and exploration factors σ_P, σ_Y . The complete implementation, including all training hyperparameters and configuration settings, is provided in the accompanying GitHub repository.

Algorithm 1 GAIL training with OpenAI-ES

Input: Policies for predator $\pi_{P,\phi,\varphi}$ and prey $\pi_{Y,\phi,\varphi}$,
Discriminators $D_{P,\beta}$ and $D_{Y,\beta}$,
Encoders $E_{P,\psi}$ and $E_{Y,\psi}$,
Expert trajectories $\tau_{E,P}$ and $\tau_{E,Y}$

Output: Best $\pi_{P,\varphi}^*$ and $\pi_{Y,\varphi}^*$

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1: Pretrain $\pi_{P_{\phi,\varphi}}$ and $\pi_{Y_{\phi,\varphi}}$ with BC
2: Train and freeze $E_{P,\psi}$ and $E_{Y,\psi}$
3: Initialize EMA policies $\bar{\pi}_{P_{\phi,\varphi}}$ and $\bar{\pi}_{Y_{\phi,\varphi}}$
4: for $i = 0, 1, 2, \dots, M$ generations do
5: Execute $\bar{\pi}_{P_{\phi,\varphi}}$ and $\bar{\pi}_{Y_{\phi,\varphi}}$ to collect rollouts
6: for $j = 1, \dots, J$ $D_{P,\beta}$ -update-ratio do
7: Update predator discriminator $D_{P,\beta}$
8: end for
9: for $j = 1, \dots, J$ $D_{Y,\beta}$ -update-ratio do
10: Update prey discriminator $D_{Y,\beta}$
11: end for
12: optimize with ES: PIN $\pi_{P_{\phi}}$ and AN $\pi_{P_{\varphi}}$
13: PIN $\pi_{Y_{\phi}}$ and AN $\pi_{Y_{\varphi}}$
14: update EMA: $\bar{\pi}_{P_{\phi,\varphi}} \leftarrow \text{EMA}(\pi_{P_{\phi,\varphi}})$
15: $\bar{\pi}_{Y_{\phi,\varphi}} \leftarrow \text{EMA}(\pi_{Y_{\phi,\varphi}})$
16: apply decay γ to: learning rates λ_P, λ_Y
17: exploration factors σ_P, σ_Y
18: end for

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### III. SYNTHETIC AND BIOLOGICAL DATA PREPARATION

We prepared two complementary predator-prey datasets for policy learning. Synthetic trajectories with predefined interaction rules provide controlled expert demonstrations for validating the role-specific architecture and co-evolutionary training procedure, while biological trajectories reconstructed from experimental recordings provide demonstrations of real predator-prey interactions. Both datasets were processed into the same state and transition representations used by the learning framework.

### A. Couzin-Based Synthetic Data

To generate the synthetic expert demonstrations, we adapted the predator-prey simulation provided by Haeri [23], which extends the original Couzin model [1] by introducing a predator (Fig. 2). Prey follow the local repulsion, orientation, and attraction rules of the Couzin model, with additional predator avoidance within a threat radius of 30 simulation units. Simulations were conducted in a  $50 \times 50$  two-dimensional arena in simulation units for 500 steps, with a constant agent speed of 5 units per time unit, a step size of 0.5, and a maximum per-step turn of 0.25 rad.

At each time step  $t$ , the predator selects the nearest prey, whose index  $i_t^*$  is given by

$$i_t^* = \arg \min_{i \in I} \|\mathbf{x}_{i,t} - \mathbf{x}_{P,t}\| \quad , \quad (4)$$

where  $\mathbf{x}_i, t \in \mathbb{R}^2$  and  $\mathbf{x}_P, t \in \mathbb{R}^2$  denote the positions of prey  $i$  and the predator, respectively [23]. The corresponding desired pursuit direction is

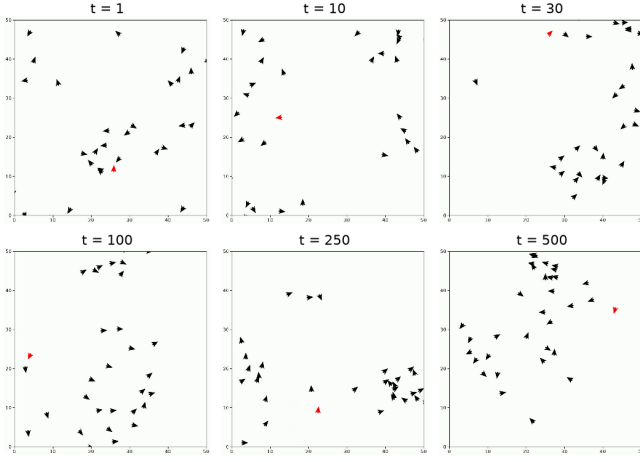


Fig. 2: Snapshots of the Cousin-based predator–prey simulation at different time steps ( $t = 1, 10, 30, 100, 250, 500$ ), illustrating the temporal evolution of the synthetic expert trajectories [1], [23].

$$d_{P,t} = \frac{\mathbf{x}_{i_t^*,t} - \mathbf{x}_{P,t}}{\|\mathbf{x}_{i_t^*,t} - \mathbf{x}_{P,t}\|}. \quad (5)$$

The predator turns toward this direction subject to the same bounded-turn dynamics as the remaining agents and subsequently moves at constant speed. Prey retain the Cousin interaction rules, augmented by predator avoidance within a predefined threat radius. Representative snapshots of the resulting synthetic predator–prey trajectories are shown in Fig. 2. These trajectories provide continuous demonstrations without tracking-induced gaps or identity ambiguities, with predefined interaction dynamics for subsequent policy learning.

### B. Biological data

The recordings capture predator–prey interactions between groups of sunbleaks (*Leucaspis delineatus*) and a northern pike (*Esox lucius*) in a controlled laboratory aquarium (Fig. 3). Trials were conducted in a featureless  $1.5 \times 1.5$  m tank, where groups of prey fish were introduced before the predator was released following an initial acclimatization phase. Visual shielding, vibration damping, and controlled environmental conditions were used to minimize external disturbances. Behavior was recorded from above using a four-camera array. The recordings had a resolution of  $2160 \times 2160$  pixels at 30 fps.

The trajectory dataset was restricted to trials with 32 and 16 prey individuals. The total duration was 16 h 42 min 49 s for the 32-prey recordings and 10 h 59 min 30 s for the 16-prey recordings. Active pursuit behavior was underrepresented: 59 successful attacks (32 for 32-prey condition and 27 for 16-prey condition) were observed, corresponding to 2 min 33.9 s of total footage.

The trajectory reconstruction pipeline consisted of two complementary components. Individual trajectories were extracted using a detection-and-tracking pipeline based on

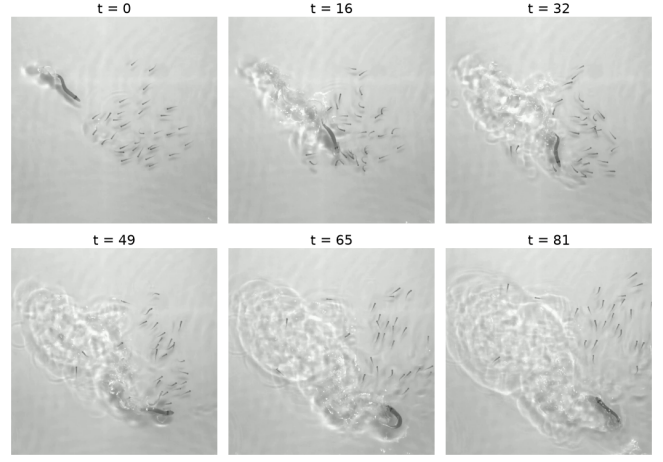


Fig. 3: Representative frames from a biological predator–prey recording, shown at six evenly spaced time points to illustrate the temporal progression of the interaction.

a fine-tuned YOLOv11 model [24] and DeepSORT [25]. YOLOv11 detected prey individuals and the predator head, while DeepSORT associated these detections across consecutive frames to preserve individual identities. We represented fish positions by the center points of their corresponding bounding boxes. Predator detection was highly stable, with a mean absolute count error of 0.03 fish per frame. Prey detection was more strongly affected by occlusions, reflections, shadows, and dense schooling behavior, resulting in a mean absolute count error of 3.11 fish per frame. We implemented an additional hand-labeling pipeline to account for tracking difficulties during attacks involving rapid motion and abrupt changes in direction. This ensured that the expert data used for training accurately represented pursuit and evasion behavior.

We filtered the reconstructed trajectories to retain only continuous sequences containing exactly one predator and either 32 or 16 prey with consistent identities. We discarded frames without predator detection, and removed implausible speed jumps caused by detection errors or identity switches. We visually inspected the remaining trajectories and divided them into fixed-length temporal windows for subsequent state construction and imitation learning. Using a window length of ten frames, this procedure yielded 3,771 valid trajectory windows (793 for 32-prey and 2,978 for 16-prey). For the biological dataset, all retained trajectory windows were used as expert demonstrations during adversarial imitation learning, without a separate recording-level hold-out set.

### C. State Representation and Tensor Construction

To apply the same learning architecture to both synthetic and biological data, we convert all trajectories into a common role-specific state representation. Following [8], [6], interactions are represented from the perspective of each focal individual, with every focal–neighbor pair described by relative position and velocity in the focal coordinate frame. To accommodate variable prey group sizes, neighbor features

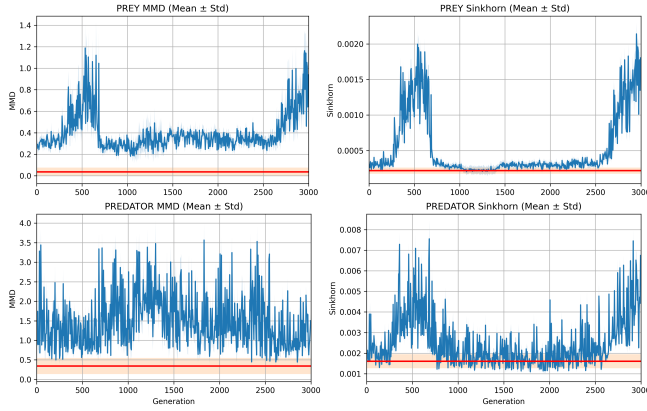


Fig. 4: MMD and Sinkhorn distances during training on Couzin demonstrations for prey (top) and predator (bottom). MMD is shown on the left and Sinkhorn distance on the right. Lower values indicate closer agreement with the expert data; red lines denote the expert–expert baseline.

are padded to 32 slots and accompanied by a binary activity mask identifying valid individuals. Consecutive states are then stacked into fixed-length trajectory windows.

This representation maps synthetic and biological trajectories to the same role-specific model interface, allowing the policy, transition encoder, and discriminator architectures to remain unchanged across synthetic validation and biological imitation learning. The two settings therefore differ only in their expert demonstrations and environment-specific motion parameters.

#### IV. RESULTS

We evaluated the main model in two stages. First, we validated the framework on synthetic predator–prey trajectories generated by the Couzin-based simulation. We then transferred it to more complex biological trajectories reconstructed from real predator–prey recordings.

##### A. Couzin-Based Predator–Prey Verification

We first evaluate whether the proposed framework can reproduce known predator–prey dynamics in the controlled Couzin setting. Training convergence is assessed using Maximum Mean Discrepancy (MMD) [26] and Sinkhorn [27] distance between expert and policy-generated trajectories. Both metrics are used because they provide complementary views of convergence: Sinkhorn compares expert and generated transitions in the learned latent representation, whereas MMD assesses distributional agreement between the resulting trajectory samples (Fig. 4). The red line indicates the Monte-Carlo expert baseline, obtained by repeatedly computing the metric between two randomly sampled expert batches. Convergence toward this baseline indicates increasingly expert-like behavior. The prey metrics closely approach the expert baseline during training, whereas the predator metrics remain substantially more volatile and generally farther from it. Accordingly, we selected the checkpoint with the lowest Sinkhorn distance as the final model.

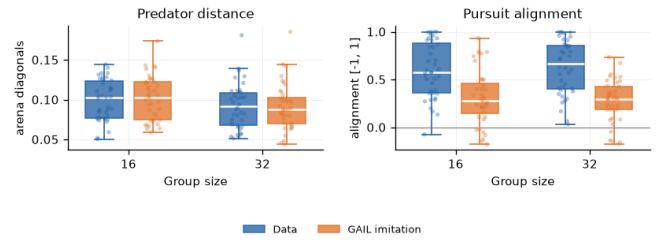


Fig. 5: Predator behavior in Couzin expert and GAIL rollouts. Boxplots show normalized predator distance (NPD; left) and pursuit alignment (PA; right) over 40 attack events for prey group sizes  $N = 16$  and  $N = 32$ . Higher PA indicates stronger alignment with the pursued prey.

We further compare Couzin expert trajectories and GAIL-generated rollouts using four behavioral metrics subsequently applied to the biological data. Predator behavior is characterized by normalized predator distance (NPD), capturing its spatial relation to the prey group, and pursuit alignment (PA), measuring directional pursuit. Prey response is quantified by escape alignment (EA) and continuous reorientation  $R$ , capturing directional avoidance and short-term changes in turning behavior, respectively. For each group size, we analyzed 40 independent attack events from held-out Couzin trajectories that were not used during training, selecting at most one event from each source trajectory. Expert and GAIL rollouts started from the same event state and were evaluated over the next 20 simulation steps to capture the immediate behavioral response while limiting accumulated rollout drift.

In Fig. 5 we compare predator-level metrics between GAIL rollouts and Couzin ground truth. NPD is nearly identical for both group sizes, with mean values of 0.100 and 0.102 for  $N = 16$ , and 0.092 and 0.091 for  $N = 32$  (Couzin vs. GAIL). PA is lower for GAIL, decreasing from 0.597 to 0.314 for  $N = 16$  and from 0.631 to 0.275 for  $N = 32$ , indicating weaker directional pursuit.

In Fig. 6 we compare prey-level responses. EA is also lower for GAIL, with mean values of 0.109 and 0.154 for  $N = 16$  and  $N = 32$ , compared with 0.223 and 0.259 for the Couzin expert. In contrast, GAIL produces stronger negative reorientation, with  $R = -0.044$  and  $-0.049$ , compared with  $-0.026$  and  $-0.033$  in the Couzin trajectories. Despite these differences in magnitude, the learned policy preserves the direction of the expert responses.

##### B. Biological Predator–Prey Imitation

Biological training shows convergence toward the expert distribution (Fig. 7). The prey Sinkhorn distance remains close to the expert baseline for extended parts of training, although MMD exhibits several pronounced deviations, while the predator metrics remain more volatile and generally farther from the baseline. The final predator and prey policies were selected based on their lowest Sinkhorn distance.

Following the Couzin verification, we apply the same framework and four behavioral metrics to biological predator–prey trajectories for  $N = 16$  and  $N = 32$ . In

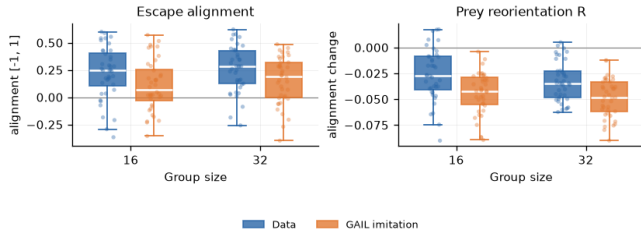


Fig. 6: Prey behavior in Couzin expert and GAIL rollouts. Boxplots show escape alignment (EA; left) and continuous reorientation ( $R$ ; right) for 40 paired attack events per prey group size ( $N = 16$  and  $N = 32$ ). Higher EA indicates stronger alignment away from the predator.

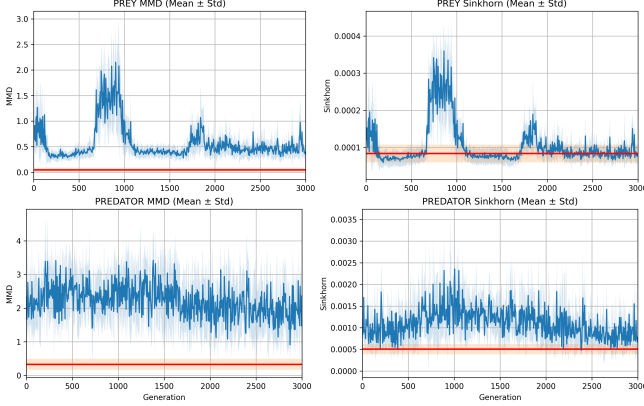


Fig. 7: MMD and Sinkhorn distances during training on biological demonstrations for prey (top) and predator (bottom). MMD is shown on the left and Sinkhorn distance on the right. Lower values indicate closer agreement with the expert data; red lines denote the expert-expert baseline.

Fig. 8 we compare predator-level metrics between biological trajectories and GAIL rollouts. NPD differs across group sizes, with mean values of 0.093 vs. 0.134 for  $N = 16$  and 0.188 vs. 0.111 for  $N = 32$  (biological vs. GAIL). PA is substantially lower in GAIL, decreasing from 0.439 to 0.009 for  $N = 16$  and from 0.073 to 0.016 for  $N = 32$ , indicating weaker learned pursuit behavior.

In Fig. 9 we compare prey-level responses. EA is also lower in GAIL, with mean values of 0.016 and  $-0.012$  for  $N = 16$  and  $N = 32$ , compared with 0.299 and 0.065 in the biological trajectories. In contrast, reorientation  $R$  is reproduced more closely, with biological means of  $-0.003$  and 0.001 compared with approximately  $-0.007$  for both GAIL conditions.

Qualitative inspection suggested that the generated rollouts retained aspects of the overall group motion, despite the weaker directional pursuit and escape responses (see accompanying video submission).

## V. DISCUSSION AND CONCLUSION

This study develops a co-evolutionary imitation-learning framework for learning predator-prey interactions directly from demonstrations using separate, interacting policies for

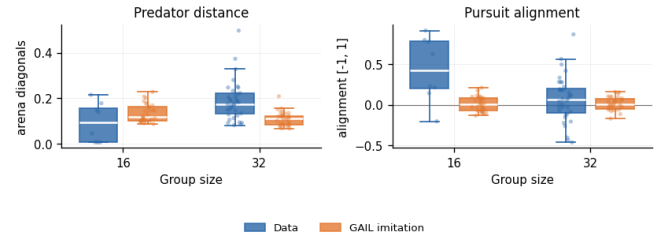


Fig. 8: Predator behavior in biological trajectories and GAIL rollouts. Boxplots show normalized predator distance (NPD; left) and pursuit alignment (PA; right) for prey group sizes  $N = 16$  and  $N = 32$ . Higher PA indicates stronger alignment with the pursued prey.

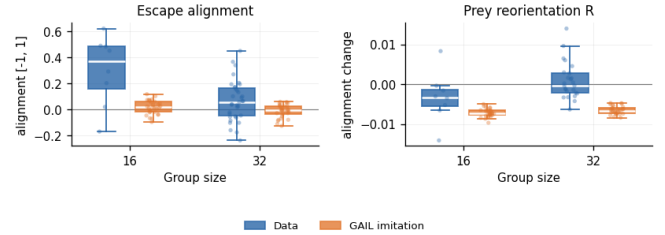


Fig. 9: Prey behavior in biological trajectories and GAIL rollouts. Boxplots show escape alignment (EA; left) and continuous reorientation ( $R$ ; right) for prey group sizes  $N = 16$  and  $N = 32$ . Higher EA indicates stronger alignment away from the predator.

predators and prey. We first validate the framework on controlled synthetic trajectories and then apply the same architecture to biological recordings. The synthetic results show that the method can capture several properties of the synthetic predator-prey dynamics, with closer agreement in collective organization, while it reproduces pursuit and escape responses less accurately. A similar but more pronounced pattern appears in biological data, where collective-level behavior is captured more closely than explicit predator pursuit and prey escape.

Our results are broadly consistent with previous adversarial imitation-learning studies while highlighting the additional difficulty of heterogeneous predator-prey interactions. Prior work has successfully reproduced homogeneous collective behaviors such as milling, schooling, and swarming using shared policies [6]. Collective Behavior Imitation Learning extends to chasing behavior but still relies on a multi-instance single-policy formulation with explicit role-dependent objectives [15]. In contrast, our framework learns separate predator and prey policies directly from demonstrations, allowing each policy to specialize in a distinct behavioral distribution rather than representing pursuit and evasion within a common controller. This role-specific decomposition is particularly useful for biological data, where predator and prey exhibit strongly asymmetric behaviors and available interaction data can be limited, because it can reduce interference between incompatible behavioral

strategies and enables their learned responses to be analyzed separately.

Our approach has several limitations. First, active pursuit and evasion events are sparse in the biological dataset, causing training to be dominated by ordinary collective motion. Second, the policies control only heading while assuming constant speed, preventing them from reproducing acceleration and rapid escape responses. Tracking errors and the restriction to two-dimensional trajectories introduce further uncertainty. As an effect, the learned policies capture aspects of the overall collective organization more closely than predator pursuit and prey escape.

Future work should therefore focus on richer biological datasets containing more frequent pursuit and evasion events and improved trajectory reconstruction to preserve longer temporal contexts. Extending the action space to include variable speed and acceleration may better capture rapid biological responses. A more tightly coupled training signal, for example through a joint discriminator or a second fine-tuning stage using information from both roles, could further strengthen predator-prey interaction learning. Overall, the results demonstrate that adversarial imitation learning can recover important components of predator-prey collective motion from demonstrations, while robust imitation of survival-driven inter-species dynamics remains an open challenge. In future applications, learning decentralized, role-specific interaction policies from biological demonstrations could support the design of multi-robot systems in which agents with asymmetric roles, such as pursuit, evasion, escort, or containment, coordinate adaptively without relying on hand-crafted interaction rules.

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